- Chemistry Extended Essay -

An investigation of the most proficient water filtration method for lead.

How effectively can chitosan chemically remove Pb from H_2O, compared to activated carbon?

Word Count : 3883

Table of Contents

1. Introduction
    - 1.1 Lead in Drinking Water
    - 1.2 Absorbents
    - 1.3 Chitosan
    - 1.4 Activated Carbon
    - 1.5 Research Question
    - 1.6 Methodology
2. Comparison - Sources of Error & Future Steps
    - 2.1 Adsorption Differences
    - 2.2 Flaws in Methodology
    - 2.3 Price
3. Conclusion
4. Works Cited

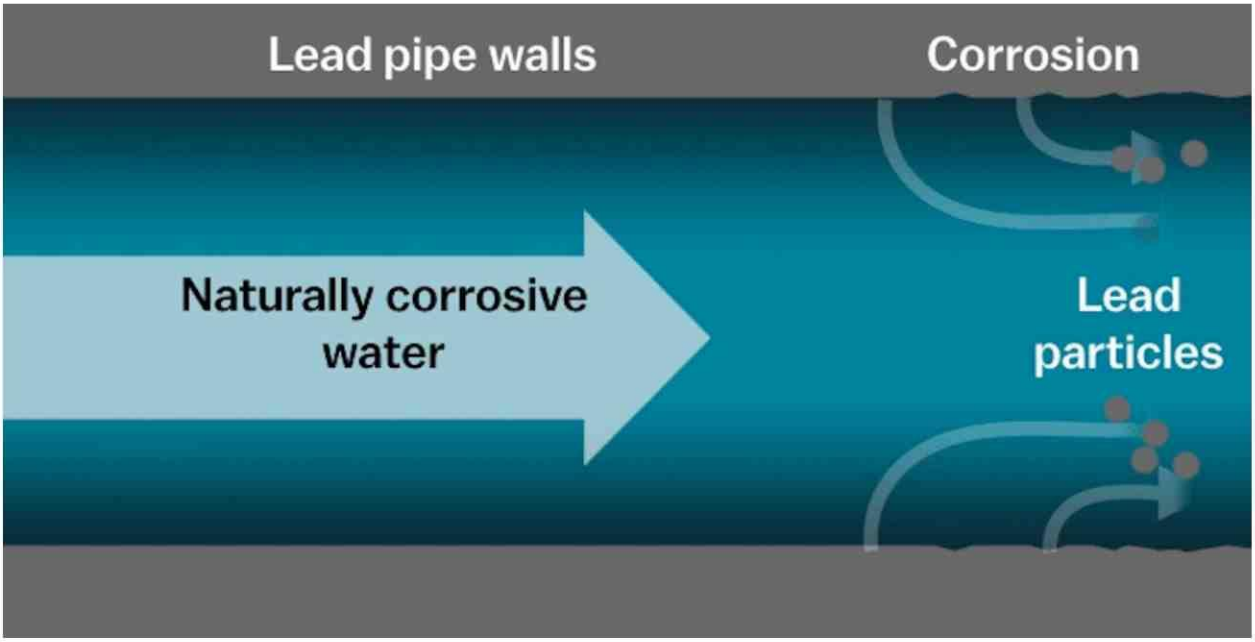
1. Introduction

1.1 - Lead in Drinking Water

Lead is one of the most common and dangerous heavy metals in the environment and is often in water systems. According to the United States Environmental Protection Agency (EPA), lead can enter drinking water when plumbing materials contain lead corrode or in other words, wear the metal down. In homes, lead pipes connecting the house to the water main are typically the most significant source of lead in the water. Lead pipes were banned in 1986, therefore they are more likely found in buildings built before that time (Denchak).

Chemically over time, corrosion occurs when lead sticks to H_2O due to water's polarity. H_2O is a polar molecule meaning that there is an unequal distribution of positive and negative charges. In water's case, the higher electronegativity of oxygen pulls the positive hydrogen ions. Therefore, H_2O is capable of adhesion, which allows the water molecules to attach to other substances, like lead; Pb^{2+} (Johnson et al). The positively charged lead ions are attracted to the strong negative oxygen from the H_2O, which results in the integration of lead into water (Helmenstine, Why is Water a Polar Molecule?). In other words, the lead will become oxidized, which damages the surface of the metal (The Electrochemical Society). Oxidation is the loss of electrons, or in Pb's case the loss of electrons to oxygen (Helmenstine, Oxidation Definition and Example in Chemistry). As oxygen is reduced or gains electrons, it forms an oxide with the metal (The Electrochemical Society), which also creates rust.

Lead Corrosion by H2O



Sourced from - Zarracina, Javier. Water Corrosion on Lead Pipes. *Vox.com*, 19 Feb. 2016, www.vox.com/2016/2/19/10972256/the-visual-guide-to-lead-poisoning. Accessed 21 Nov. 2022.

The National Institute for Occupational Safety and Health states that adults can experience cardiovascular effects, increased blood pressure, and incidence of hypertension. Along with decreased kidney function and reproductive issues (in both men and women). While children can develop behavior and learning problems, lower IQ, hyperactivity, slowed growth, hearing problems, and anemia. However, in some rare cases, ingestion of lead can cause seizures, coma, and even death.

Even a tiny amount of lead contamination is detrimental to one's health. The EPA decided the maximum contaminant level for lead in drinking water is zero since it is a

toxic metal even at low exposure levels, especially in children. Lead is persistent, and it can build up in the body over time.

Unfortunately, a significant amount of lead was found to be leaked into Flint, Michigan's water supply in 2014. The contamination resulted in the Flint residents consuming lead water for 18 months and experiencing degrading health. Not to mention, the children, city-wide, were found to have doubled or even tripled blood lead levels, endangering their livelihood (Denchak). The severity of this incident has since sparked a national outrage at the prevalence of lead piping and lack of filtration.

1.2 - Adsorbents

Many different methods of removing heavy metals from water are studied, such as "ion exchange, chemical precipitation, filtration, electrolytic processes, solvent extraction, adsorption, and biosorption" (Magdalena Wołowiec et al). Ion exchange is a chemical reaction that trades ions from the wastewater solution with one that has a similar charge (Dahmen). Next, chemical precipitation is the creation of solid, removable material from a solution by changing the insolubility (Editors of Encyclopedia Britannica). Meanwhile, filtration is the separation of solid from a liquid or a larger molecule from a smaller one (Schaller 315). Electrolytic processes, also known as electrolysis, decomposes ionic compounds by sending an electric current throughout the solution to make it lose or gain electrons (Aakash BYJU'S). Solvent extraction is the separation of compounds based on their solubility in two unmixable solutions (Chatterjee). Then, biosorption is the binding between metal ions and the surface of a

biological absorbent (Zainith et al). Finally, adsorption is the adhesion of atoms to the surface of a solid material (Helmenstine, What Adsorption Means in Chemistry).

In adsorption, the material adsorbing is called an adsorbent, while the substance being adsorbed is called the adsorbate. Adsorption is caused by Van Der Waal and electrostatic forces; attraction of molecules, between adsorbate molecules and the adsorbent's surface atoms. Therefore, the reliability of adsorbents are sorted by the amount of surface area and polarity (The International Adsorption Society). Increasing the surface area of a material allows a greater number of reactions to take place. In this case, adsobents' pore size is their surface area, the smaller the pore size the higher the internal surface area (Mercury Adsorbents). Therefore, creating a faster rate and capacity of adsorption. Meanwhile a greater positive or negative pull in a polar molecule, will increase the attraction and therefore raise the adsorbents' ability.

Adsorbents are able to absorb moisture, so they are known for their aptitude to reduce corrosion. Additionally, they have a low manufacturing price, strong regenerative traits, no sludge, are efficient, and generally uncomplicated (Diana Alonso-Segura et al., 1).

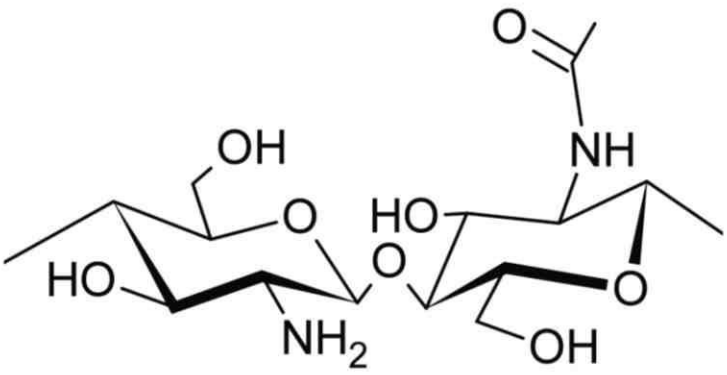
My analysis will focus on the effectiveness of adsorbents, specifically chitosan and activated carbon.

1.3 - Chitosan

According to WebMD, Chitosan, $C_{56}H_{103}N_9O_{39}$ (PubChem), is a polysaccharide or sugar derived from the chitin shells of crustaceans. Chitosan's ample hydroxyl and

amino groups allow it to attach to heavy metal ions, like lead (Yuzhe Zhang et al). A hydroxyl group is an oxygen atom with two lone pairs of electrons bonded to a hydrogen atom (Klecker et al). An amino group is made up of a nitrogen atom and two hydrogen atoms bonded together (Teeter). Just like H2O, a hydroxyl group is polar with the oxygen atom having a strong negative charge, therefore being able to attract the positive metal ions. Simularily, the nitrogen in the amino group is more electronegative than hydrogen, resulting in slight polarity.

Chitosan Molecular Structure



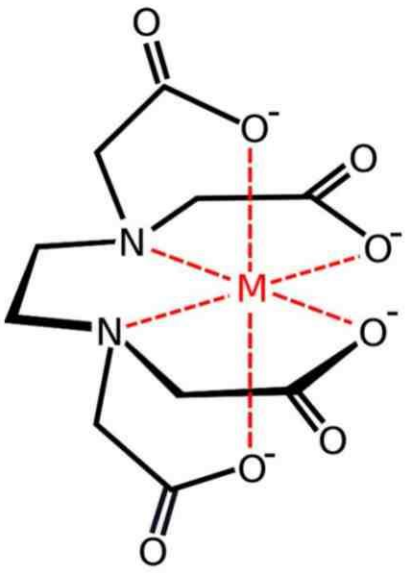
Sourced from - Teixeira-Costa, Bárbara. "Chemical Structure of Chitosan." *ResearchGate*, Oct. 2021, www.researchgate.net/publication/355787682_Chitosan_as_a_Valuable_Biomolecule_from_Seafood_Industry_Waste_in_the_Design_of_Green_Food_Packaging.

However, chitosan is capable of stabilizing in acidic environments, has a low surface area, and inadequate mechanical and thermal traits, compared to other adsorbents,

ultimately limiting its volatility (Alonso-Segura et al., 1). Therefore, studies have been conducted to modify chitosan's composition to enhance its effectiveness.

More specifically, chitosan's ability as a natural polymer ligand is altered (Pestov et al). Polymers are long, repeating chains of molecules (Bradford 1). A ligand is any molecule or atom with available lone pair electrons that irreversibly binds to a central metal ion (ScienceDirect, Ligand). Therefore, removing the metal, or lead ion from the water. However, the most common method of modifying the chitosan is by covalent crosslinking of polymers (Alonso-Segura et al., 1). The crosslinking creates a formation of polysaccharide chains and the crosslinking agent that is connected by covalent bonds, which strengthen the rigidity and hardness of the material (Vitz et al.; Jóźwiak et al.). Crosslinking agents, also known as, cross linkers are molecules that at minimum have two functional groups capable of attaching to the polymer (Jóźwiak et al). In other words, crosslinkers act as the glue for the chitosan and covalent bonding.

Ligands Bonding to Metal Ion



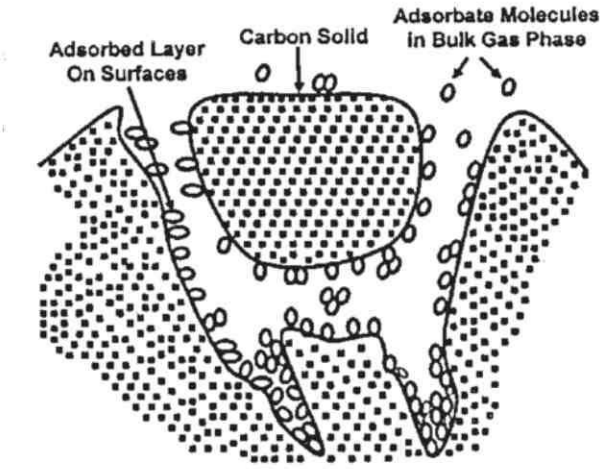
Sourced from - "Metal–EDTA Complex, Wherein the Aminocarboxylate Is a Hexadentate (Chelating) Ligand." *Wikipedia*, Wikimedia, 22 Apr. 2010, en.wikipedia.org/wiki/Ligand#/media/File:Metal-EDTA.svg. Accessed 17 Nov. 2022.

1.3 - Activated Carbon

Activated carbon, C (PubChem), is a form of graphite that is known for its high adsorption capacity and rate. Drinking water treatment systems often include activated carbon filters (Carbon Activated Corp). The graphite structure creates a high surface area, allowing the activated carbon to be capable of adsorbing various contaminants (Chemviron). Typically, the activated carbon is turned into a granular or smaller form to increase the surface area (TIGG). Additionally, activated carbon is made out of organic materials, such as coconut shells, wood, and more (Chemviron). Depending on the chosen material, characteristics and abilities can greatly differ.

Like all adsorbents, activated carbon's main source of attraction are the Van Der Waals Forces or more specifically, the London Dispersion Forces (LDF). The LDF is a temporary dipole or partly negative and positive charge, which will induce an attraction between molecules (Purdue). Though the LDF is considered the weakest intermolecular force, due to activated carbon's vast surface area, it has a relatively high adsorption ability.

Diagram of Carbon Adsorbing Particles



Sourced from - Manocha, Satish. Adsorption processes in activated carbons: Transfer of adsorbate molecules to adsorbent. *Research Gate*, www.researchgate.net/figure/Adsorption-processes-in-activated-carbons-Transfer-of-adsorbate-molecules-to-adsorbent_fig1_225940195. Accessed 21 Nov. 2022.

Similarly to chitosan, activated carbon is often modified to increase its adsorption properties of heavy metals. Polypyrrole (PPy), C4H5N (Chemical Book), is a polymer conductor that can consist of numerous substances, like chitosan. The label conductor refers to the ability to conduct charge (ScienceDirect, Conducting Polymer). The PPy nitrogen inflicts a positive charge, which enhances the adsorption rate through electrostatic interaction and ion exchange (Ray et al).

1.5 - Research Question

Chitosan is a relatively new adsorbent that is still being analyzed and tested, while activated carbon is the most commonly utilized in water filtration. Both adsorbents have the capabilities of being effective ways to clear water contamination, like lead. Therefore, *how effectively can chitosan chemically remove Pb from H_2O, compared to activated carbon?*

1.6 - Methodology

I will be comparing and contrasting two studies that focus on modified chitosan and activated carbon in the removal of lead. The focus will be on the percentage adsorption, composition of the altered materials, and the limitations of my analysis. An experiment was not able to be conducted due to the expensive price of the adsorbents, and lack of information regarding the best method to utilize raw materials.

The chitosan study, *"The Development and Characterization of a Cotton–Chitosan Composite for Lead Removal from Water"(Diana Alonso-Segura et al., 1)*, focuses on the adsorption ability of chitosan covalently crosslinked into cotton textiles. Cotton is a polysaccharide, just like chitosan, therefore the adsorbent qualities will help the composite's ability to remove lead from H2O. The ultimate goal of the study is to find an efficient biodegradable and environmentally friendly adsorbent.

The activated carbon study, *"Efficient Adsorption of Lead (II) from Aqueous Phase Solutions Using Polypyrrole-Based Activated Carbon" (Alghamdi et al.)*, details the creation of polypyrrole-based activated carbon and its capabilities in the adsorption of lead in water.

2. Comparison - Sources of Error & Future Steps

2.1 - Adsorption Differences

For the purpose of this analysis, the best performing chitosan composite, Composite 4 was chosen to compare with the activated carbon material. Citric acid (CA) and butanetetracarboxylic acid (BTCA) were used as cross-linkers, while NaH_2PO_4 acted as a catalyst.

Table 1 - Composite Formulation

Composite	H_2O_2 (35% *v/v*)	CA	BTCA	NaH_2PO_4
1	40 min	4% w/v	-	-
2	40 min	-	4% w/v	-
3	40 min	4% w/v	-	2.3% w/v
4	40 min	-	4% w/v	2.3% w/v

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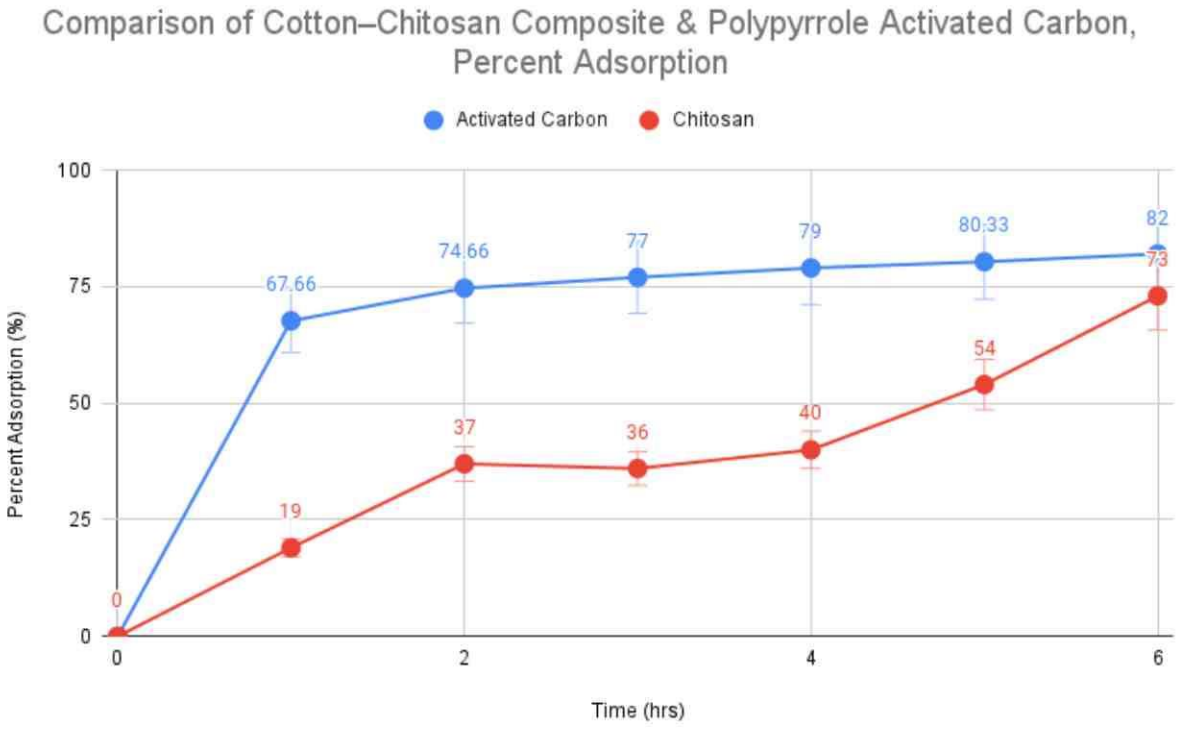
Alonso-Segura, Diana, et al. "The Development and Characterization of a Cotton–Chitosan Composite for Lead Removal from Water." *Polymers (Basel)*, 23 June 2021, pp. 1-16. *National Library of Medicine*, https://doi.org/10.3390/polym13132066. Accessed 25 Sept. 2022.

Composite 4 was found to be the most effective due to the submersion in BTCA and NaH_2PO_4 . As mentioned previously, NaH_2PO_4 is a catalyst; “a substance that speeds up a chemical reaction” (Energy.gov). Therefore, drastically improving the adsorption ability

of the composite. However, it is not known why BTCA is capable of a better result than CA, so further research could be done.

On the other hand, the mean percent adsorption for the polypyrrole activated carbon was calculated for each point, due to the similarity of each amount for the different concentrations of lead. Percent adsorption was chosen to measure the ability of adsorption because of the consistency of both material tests.

Figure 1



By 6 hours, the activated carbon has a percent adsorption of 82% and the chitosan composite at 73%; a difference of 9%. Chitosan begins to quicken in adsorption at 4 hrs, but ultimately still falls short of the activated carbon. The time needed for the activated

carbon to fully adsorb can be cut shorter than the chitosan, as it increases and stabilizes at a faster rate. The cotton-chitosan material is also noted to have “no increases after 6h” *(Diana Alonso-Segura et al., 6).* Therefore, the best adsorption would be the polypyrrole activated carbon. Data points were extracted from the studies’ section on percent adsorption, as seen in the table below:

Table 2 - Percent Adsorption Comparison of (PPy) Activated Carbon and Corron-Chitosan Composite

Time (hrs)	Polypyrrole-Based Activated Carbon Adsorption Percent (%)	Polypyrrole-Based Activated Carbon **Mean** Adsorption Percent (%)	Cotton-Chitosan Composite Adsorption Percent (%)
0	0 (all)	0	0
1	71 (10 ppm) 68 (50 ppm) 64 (100 ppm)	67.66	19
2	78 (10 ppm) 75 (50 ppm) 71(100 ppm)	74.66	37
3	79 (10 ppm) 77 (50 ppm) 75 (100 ppm)	77	36
4	80 (10 ppm) 79 (50 ppm) 78 (100 ppm)	79	40
5	82 (10 ppm) 80 (50 ppm) 79 (100 ppm)	80.33	54
6	84 (10 ppm)	82	73

	82 (50 ppm) 80 (100 ppm)		

Data sourced from :

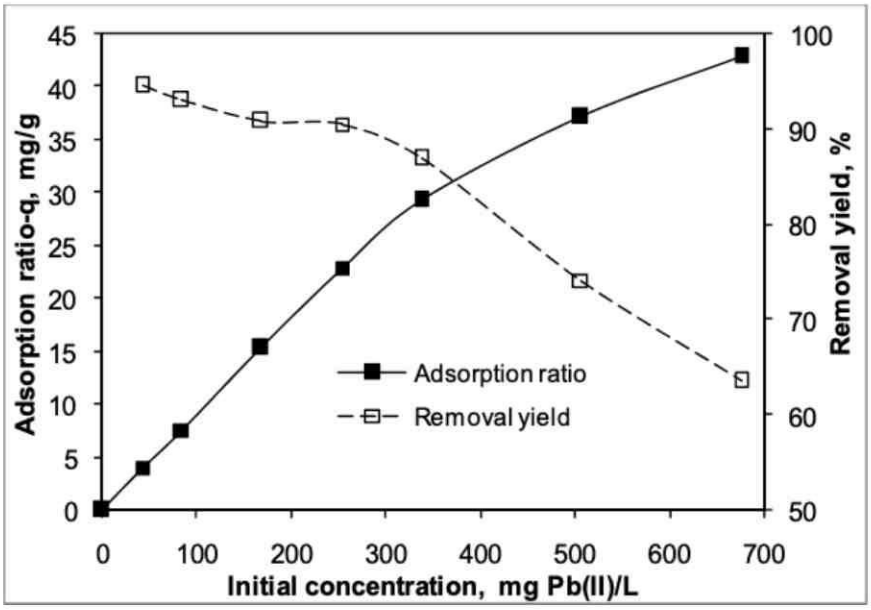
Alonso-Segura, Diana, et al. "The Development and Characterization of a Cotton–Chitosan Composite for Lead Removal from Water." *Polymers (Basel)*, 23 June 2021, pp. 1-16. *National Library of Medicine*, https://doi.org/10.3390/polym13132066. Accessed 25 Sept. 2022.

Alghamdi, Abdulaziz Ali, et al. "Efficient Adsorption of Lead (II) from Aqueous Phase Solutions Using Polypyrrole-Based Activated Carbon." *Materials (Basel)*, 24 June 2019, pp. 1-16. *National Library of Medicine*, https://doi.org/10.3390/ma12122020. Accessed 25 Sept. 2022.

For the polypyrrole-based activated carbon, the adsorption percent increases the lower the concentration of lead. The chitosan study does not test multiple concentrations of lead, however, inferences can be made by observing past experiments on the basic adsorption characteristics of pure chitosan.

The study, *"Lead (II) Removal from Aqueous Solutions by Adsorption onto Chitosan" (Asandei et al.)* observes the correlation between concentration of Pb2+ and the percent removal yield. Percent removal yield of Pb2+ is equivalent to the percent adsorption of Pb2+, just worded differently.

Figure 2 - Adsorption Ratio & Removal Yield of Pb (II) Dependent on the Initial Concentration of Pb (II) Solution



Sourced from : Asandei, Doina, et al. "Lead (II) Removal from Aqueous Solutions by Adsorption onto Chitosan." *Cellulose Chemistry and Technology*, 26 Feb. 2009. PDF.

The percent adsorption decreases with the increase in lead ion concentration. Therefore, chitosan has the same disproportional relationship between percent adsorption and concentration of Pb2+ as the activated carbon material. Inferring that the chitosan composite will have a similar reaction. Since, both adsorbents both display the same pattern of behaviors in various concentrations of the adsorbate, the comparison is further supported. The overall percentage adsorbed from the solution decreases with the more lead ions because the material runs out of surface area to attract the Pb2+.

Both studies also utilized the same formulas to calculate the adsorbate's ability; adsorption capacity (1) and percent adsorption (2) :

$$q_e = \frac{(C_0 - C_e)V}{m} \qquad (1)$$

$$\%\, Absorption = \frac{(C_0 - C_e)}{C_0}(100) \qquad (2)$$

Concerning the chitosan study, the adsorption capacity is q_e (mg/g). The C_0 and C_e (mg/L) stands for the initial and equilibrium concentrations of lead. V (L) is the volume of solution. Then, m (g) is the adsorbents' mass of the composite *(Diana Alonso-Segura et al., 3)*.

While for the activated carbon experiments, instead, q_t(mg/g) is the adsorption capacity (3) of PPyAC4 (polypyrrole-activated carbon) for Pb2+ at time *t* (min); not equilibrium *(Alghamdi et al)*. However, the percent adsorption formulas are identical.

$$q_t = \frac{(C_0 - C_e)V}{m} \qquad (3)$$

C_0 is the liquid phase concentration of Pb2+ before adsorption. C_e (mg/L) is the liquid phase concentration of Pb2+ after adsorption. V (L) is the volume of the adsorption

solution. Finally, *m* (g) is the mass of the dry adsorbent *(Alghamdi et al)*. Due to the identical usage of equations, the comparison of both results are more credible since there is less possibility of error.

Finally, both adsorption experiments utilize $Pb(NO_3)_2$, which means that the sources of Pb are the same; limiting the amount of variables that could influence the comparison of the two studies. However, while the chitosan composite is directly submerged in $Pb(NO_3)_2$; the PPyAC4 is placed in a stock solution of 500 ppm, Pb2+ by dissolving $Pb(NO_3)_2$ in deionized water *(Alghamdi et al)*. Therefore, there is a difference in the environments, which leaves room for error. Although the original adsorbate is the same, $Pb(NO_3)_2$; the process in which the Pb is extracted differs.

2.2 - Flaws in Methodology

The concentration of lead tested with the opposing materials were scrutinized. The polypyrrole activated carbon tested at 10 ppm, 50 ppm, and 100 ppm, which was then averaged to 53.3 ppm. However, the cotton chitosan composite was analyzed in 100 mL of Pb solution at a concentration of 40 mg/L. Therefore the mg/L needs to be converted into parts per million.

$$100\ mL \text{ of Pb Solution}$$

$$\text{Solution} : 40\ mg/L \rightarrow 0.040\ g/L \rightarrow\ 0.004\ g$$

$$\left(\frac{0.004g\ Pb}{100g\ H_2O}\right)(1,000,000)\ =\ 40\ ppm$$

Therefore, the activated carbon is being tested at a higher concentration of Pb, 53.3 ppm, than the chitosan composite, 40 ppm. As analyzed prior, the higher concentration the lower percent adsorption. If the lead concentration of the chitosan composite was also tested at 53.3ppm, then the percent adsorption would be even less. Activated carbon's adsorption is more reliable and effective than the cotton-chitosan composite.

The general form of the adsorbents noticeably differ, as the polypyrrole-based activated carbon is in granules versus the solid cotton-chitosan textiles. Granular activated carbon is smaller, and is in an increased quantity, which allows a greater number of reactions to take place. However, since activated carbon has been determined to be the best adsorbent, based solely on the given study data already, there is a way that the chitosan composite can be improved.

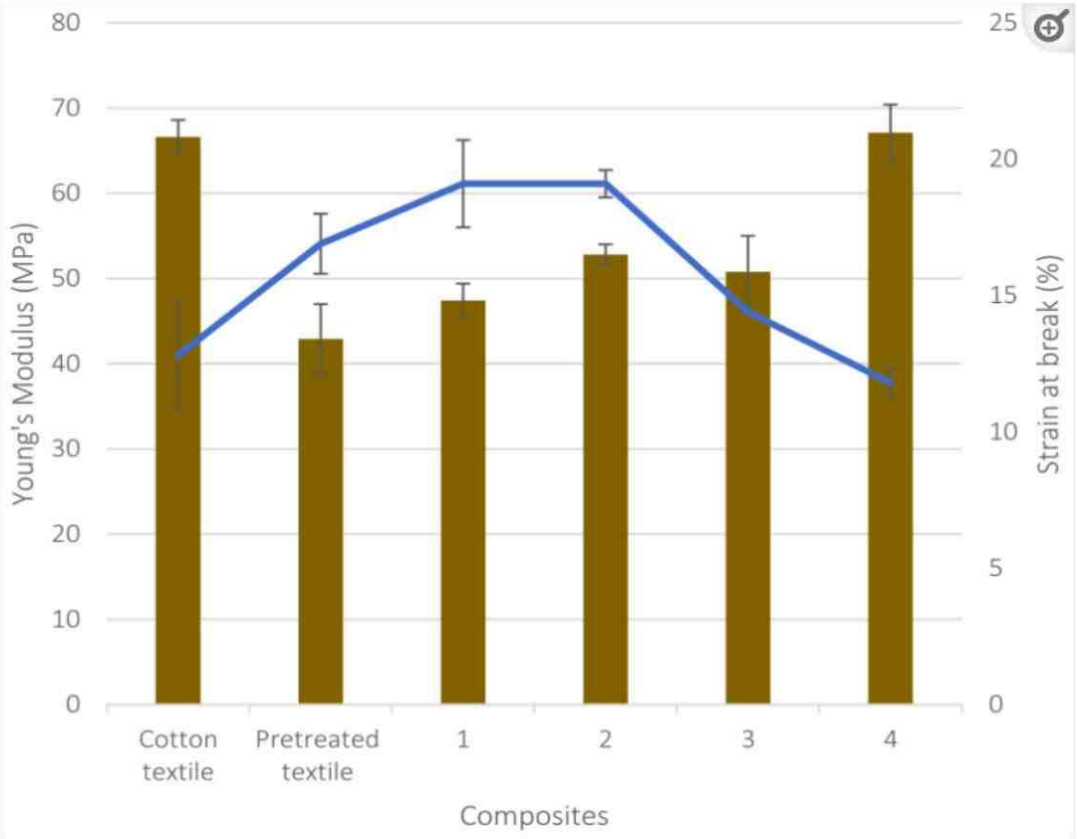
Table 3 - Chitosan Crosslinked into the Composites Developed Herein

Composite	$mg_{chitosan}$ to $g_{composite}$ *
1	13.56 ± 2.00
2	22.60 ± 1.81
3	20.13 ± 0.87
4	27.62 ± 5.03

Sourced from : Alonso-Segura, Diana, et al. "The Development and Characterization of a Cotton–Chitosan Composite for Lead Removal from Water." *Polymers (Basel)*, 23 June 2021, pp. 1-16. *National Library of Medicine*, https://doi.org/10.3390/polym13132066. Accessed 25 Sept. 2022.

Composite 4 has the best adsorption performance, and has the highest amount of chitosan crosslinked into the cotton. Therefore, if more chitosan is incorporated into the composite, then it will have a higher adsorption percentage. Now there is a worry that the composite will be too delicate, however a tensile strength test was performed.

Figure 3 - Average Tensile Strength and Breaking Point Strain of Cotton Textiles, Pretreated Textiles with H2O2, and Composites



Sourced from : Alonso-Segura, Diana, et al. "The Development and Characterization of a Cotton–Chitosan Composite for Lead Removal from Water." *Polymers (Basel)*, 23 June 2021, pp. 1-16. *National Library of Medicine*, https://doi.org/10.3390/polym13132066. Accessed 25 Sept. 2022.

The graph measures in both Young's modulus and percent strain at break. Young's modulus is a "measure of a material's capacity to withstand changes in length when under lengthwise tension or compression" (Encyclopedia Britannica). Composite 4 has the highest strain at the breaking point and, therefore, the composite with the highest value.

An increased incorporation of chitosan within the composite both raises the adsorption percentage and strength of the fabric. If further tests were conducted, but with a $mg_{chitosan}$ to $g_{composite}$ higher than 27.62 ± 5.03, then the adsorption percentage could exceed activated carbon's capabilities.

2.3 - Price

However, although from the studies chosen, activated carbon is considered the best Pb absorbent option, factors aside from the actual absorption rate can change the answer. Such as the cost of production and affordability for each substance. The creation process for activated carbon and the chitosan textiles took more than one day, but the cost of the materials and machinery required was never listed. Therefore, if activated carbon was exponentially more expensive than chitosan; chitosan would be the optimal choice opposed to activated carbon. Cheaper prices allow the absorbent to be accessible to a broader range of consumers and better assist the decontamination of Pb in H2O.

Notably, both studies are from completely different parts of the world. The activated carbon study is from Saudi Arabia and the composite chitosan is conducted in

Mexico. Prices would even more so differ from one another due to accessibility and shipping of resources.

The amount of time should also come into consideration. In general, both procedures to prepare the absorbents are tedious. They both include submersion in solution, washing with distilled water, and heating in the furnace. However, the amount of time allotted for all the steps to be effectively completed is much longer for the creation of the Polypyrrole-Based Activated Carbon. The activated carbon component took around 72 hrs to finish, while the chitosan composite is a little longer than 48 hrs. Therefore, the activated carbon procedure seems to be more labor-intensive than when creating the chitosan composite.

However, it should be noted that in the activated carbon study, they used 125 mg of PPyAC4 (activated carbon material), while the chitosan worked with 250 mg of dried composites. In order for the chitosan composite to effectively adsorb lead at 73%, it needs two times the amount of material compared to activated carbon. Though the creation time of the chitosan composite is longer, more needs to be made to be useful.

Therefore, if the cumulative cost of each study is revealed, it could potentially cause the better absorbent to be the cotton-chitosan composite instead of activated carbon. Comparing the lengthy time-period and effort required to produce the material could influence the determination of the better adsorbent.

3. Conclusion

By generally analyzing the two studies, the best lead adsorbent is polypyrrole-based activated carbon with percent adsorption of 9% higher than the chitosan-cotton composite. They utilize the same formulas to calculate the percent adsorption, making them effective comparisons.

However, there are sources of error between both studies as though the lead originates from the same solution, Pb(NO3)2, the methods of extracting the adsorbate differ. Additionally, they were tested in different concentrations of lead. But, both materials' percent adsorption was observed or inferred to decrease with an increase in Pb concentration. Therefore, with the unique composites for the studies, the activated carbon would still be superior even if the Pb concentration were to increase.

There are still opportunities for the chitosan-cotton composite to be the more effective adsorbent. If the chitosan-cotton composite has a mgchitosan to gcomposite higher than 27.62 ± 5.03, the percent adsorption capabilities and durability will improve. Determining the better adsorbent is more than just the ability to adsorb, but also accessibility. The production process is held in different countries, which can affect the cost. Chitosan may be a cheaper alternative to activated carbon in those countries, not to mention the composite does have a 24-hour production time than the activated carbon. The cotton-chitosan composite will exceed the polypyrrole-based activated carbon if it is cheaper.

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